RAFFLES INSTITUTION

2024 YEAR 6 TERM 3 TIMED PRACTICE

27 June 2024 Time: 3 hours

H2 PHYSICS

Section A

INSTRUCTIONS TO CANDIDATES

There are 2 sections in this paper.

Section A consists of **30** multiple-choice questions. For each question, four suggested answers are given. You are to choose the most appropriate one and shade your answer on the OMR form, using a soft pencil.

You are advised to attempt Section A first. The **OMR form will be collected at the end of the first 1 hour**.

Section B consists of **8** structured questions. You are to write your answers in the spaces provided.

Attempt ALL questions in Sections A and B.

This document consists of **16** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
		=	$(1/(36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^{2} kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	ρgh
gravitational potential	ϕ	=	$-Gm/r$
temperature	T/K	=	T / °C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas molecule	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{x_0^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$1/R$	=	$1/R_1 + 1/R_2 + \ldots$
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
radioactive decay	x	=	$x_0 \exp(-\lambda t)$
decay constant	λ	=	$\ln 2 / t_{1/2}$

Section A (30 marks)

1 A meteorite falls freely towards the centre of an isolated spherical planet of radius R. When it reaches the surface of the planet, its acceleration is g_s.

What is its acceleration when at a height $2R$ above the surface?

A $\frac{1}{9}g_s$ **B** $\frac{1}{4}g_s$ **C** $\frac{1}{3}g_s$ **D** $\frac{1}{2}g_s$

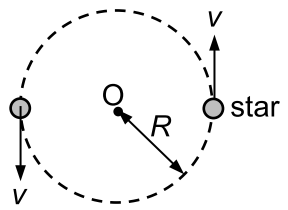
2 A satellite orbits a planet at a distance r from its centre. Its gravitational potential energy is -3.2 MJ.

Another identical satellite orbits the planet at a distance $2r$ from its centre.

What is the sum of the kinetic energy and the gravitational potential energy of this second satellite?

A –0.40 MJ **B** –0.80 MJ **C** –1.6 MJ **D** –6.4 MJ

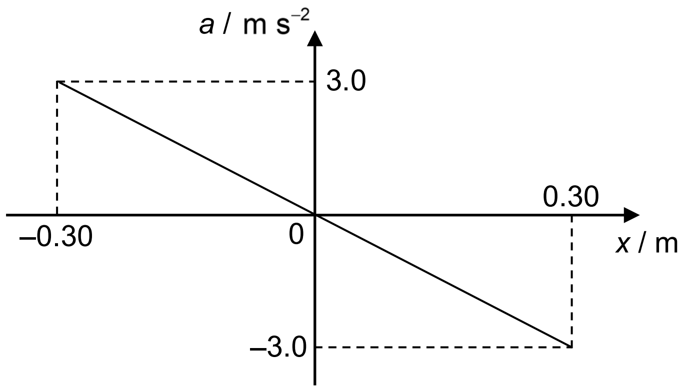
3 Two stars of equal mass M move with constant speed v in a circular orbit of radius R about their common centre of mass O.



What is the net force on each star?

A $\dfrac{GM^2}{4R^2}$ **B** $\dfrac{2Mv^2}{R}$ **C** $\dfrac{Mv^2}{2R}$ **D** $\dfrac{GM^2}{R^2}$

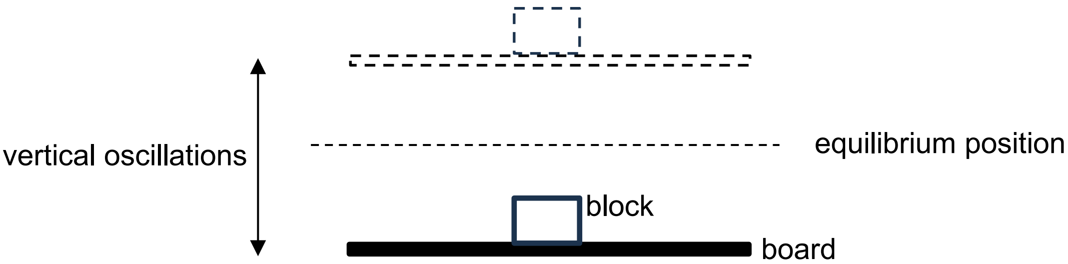
4 The graph shows the variation with displacement x from the equilibrium position of the acceleration a of an object undergoing simple harmonic motion.



What is the speed of the object when $x = 0.10$ m ?

A 0.32 m s^{-1} **B** 0.80 m s^{-1} **C** 0.89 m s^{-1} **D** 2.8 m s^{-1}

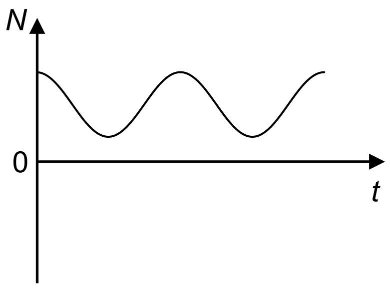
5 A horizontal board, with a small block placed on top of it, is oscillating vertically in simple harmonic motion. The block remains in contact with the board during the oscillations.



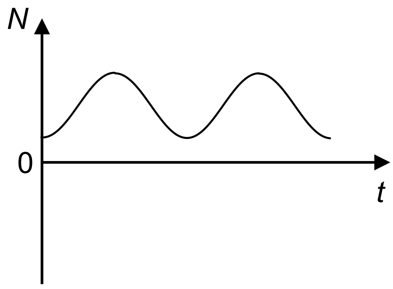
At time $t = 0$, the board is at the lowest point of its oscillation below its equilibrium position.

Which graph shows the variation with time t of the normal contact force N between the block and the board?

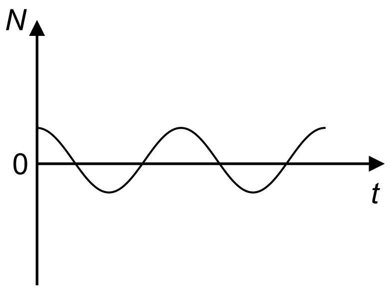
A



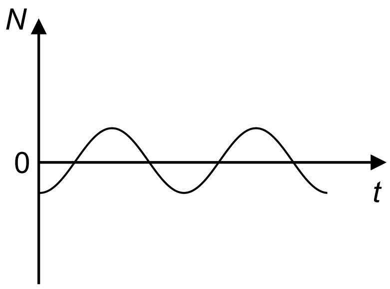
B



C

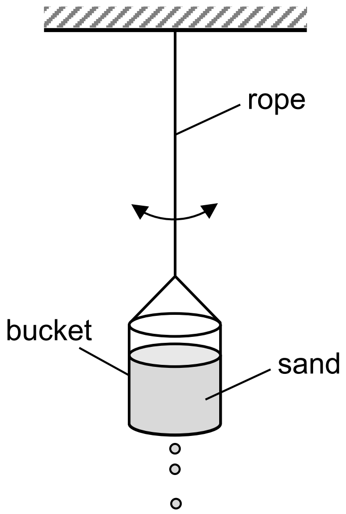


D



6 A bucket completely filled with sand is suspended from a fixed point by a light inelastic rope. It is given a small horizontal displacement from its rest position.

When the bucket is released, it oscillates as a pendulum in simple harmonic motion.



However, there is a small hole at the bottom of the bucket, which allows sand to slowly trickle out of the bucket.

How would the period of the oscillations of the bucket vary with time as sand trickles out until the bucket is empty?

A The period of oscillation remains constant with time.

B The period of oscillation increases with time.

C The period of oscillation decreases with time.

D The period of oscillation increases, then decreases back to its original value.

7 A sound wave of frequency 540 Hz travels in a gas at a speed of 340 m s^{-1}.

What is the distance between two points if the phase difference between them in the direction of the wave travel is $\frac{3}{4}\pi$ rad?

A 0.24 m **B** 0.47 m **C** 0.60 m **D** 1.2 m

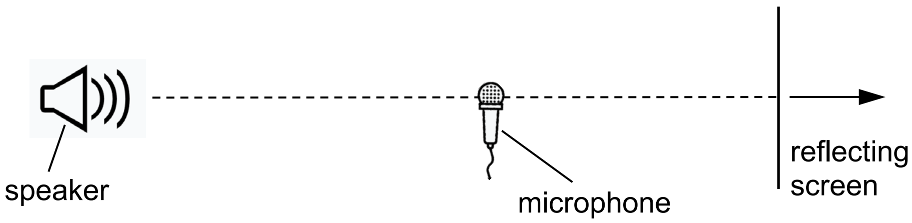
8 A plane wave of amplitude a is incident normally on a surface of area S. The energy per unit time intercepted by the surface is P.

The amplitude of the wave is increased to $4a$ and the area of the surface is reduced to $\frac{S}{4}$.

How much energy per unit time is intercepted by this smaller surface?

A P **B** $4P$ **C** $16P$ **D** $64P$

9 A microphone is kept stationary between a speaker emitting sound waves at a single fixed frequency and a reflecting screen. The speaker, microphone and reflecting screen are arranged along the same horizontal line.



A maximum signal is detected by the microphone. As the reflecting screen is moved through a distance of 80 cm away from the microphone, another six maximum signals are detected.

If the speed of sound in air is 330 m s^{-1}, what is the frequency of the sound being emitted by the speaker?

A 620 Hz **B** 1000 Hz **C** 1200 Hz **D** 2500 Hz

10 The International Space Station (ISS) orbits Earth at an altitude of 400 km. Two monochromatic lights on the surface of Earth are separated by a distance d.

Assume the pupil of a human eye has a slit width of 4.0 mm.

What is the minimum value of d for an astronaut inside the ISS to resolve the monochromatic lights with her naked eyes?

A 40 m **B** 70 m **C** 80 m **D** 140 m

11 Light of frequency 550 THz is incident normally on a diffraction grating. The angle between the two third-order diffraction maxima is $82°$.

What is the number of lines per millimetre of the grating?

A 2.5×10^{-6} **B** 4.0×10^{2} **C** 6.1×10^{2} **D** 4.0×10^{5}

12 A solar furnace made from a concave mirror of area 0.40 m^2 is used to heat water. The radiant energy from the Sun arrives at the mirror's surface at a rate of 1400 W m^{-2}. The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

What is the shortest time needed to heat 1.0 kg of water from 20 °C to 50 °C?

A 0.60 min **B** 3.8 min **C** 36 min **D** 230 min

13 A fixed mass of gas in a thermally insulated container is compressed slowly.

Which statement explains the effect on the temperature of the gas after compression?

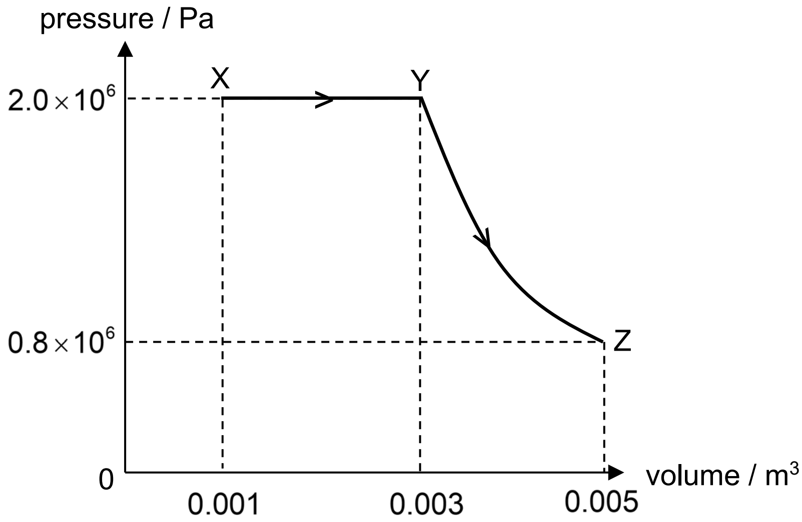
A The temperature of the gas remains constant since the compression takes place very slowly.

B The temperature of the gas remains constant since the container is thermally insulated ensuring there is no change in the internal energy of the gas.

C The temperature of the gas increases since there are more intermolecular collisions which produce more thermal energy.

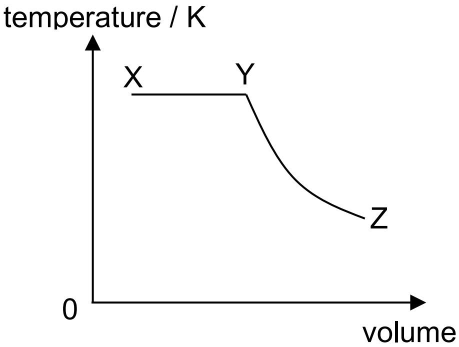
D The temperature of the gas increases since work done on the gas increases the kinetic energy of the molecules.

14 The graph shows the changes in pressure and volume of a fixed mass of ideal gas as it goes from states X to Y and to Z.

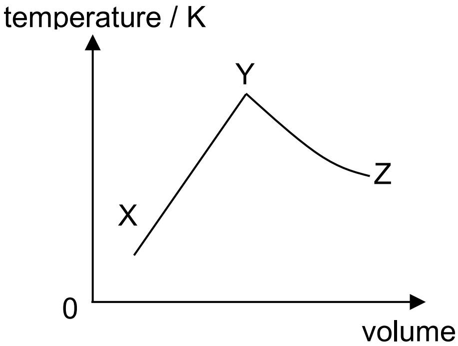


Which graph shows the corresponding changes in the temperature of the gas with volume?

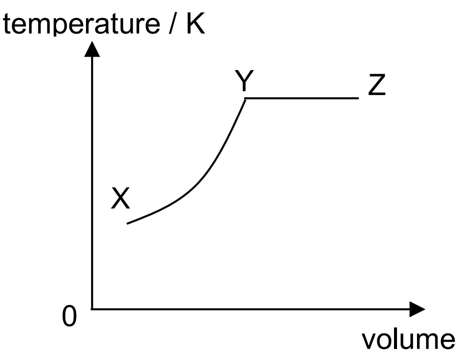
A



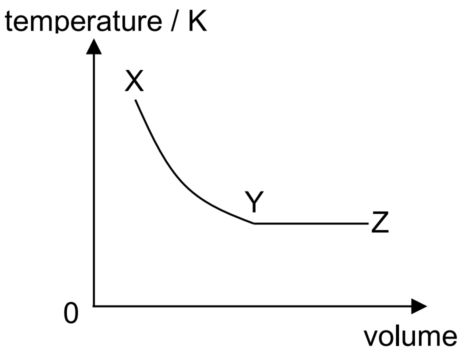
B



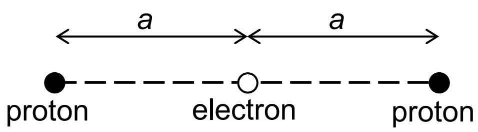
C



D



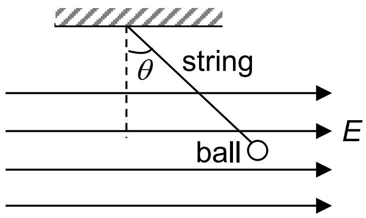
15 Two protons and an electron are placed in a straight line. The distance between the electron and each proton is *a*.



What is the work done by an external force in arranging this system of charges?

A $\dfrac{-e^2}{2\pi\varepsilon_0 a}$ **B** $\dfrac{-3e^2}{8\pi\varepsilon_0 a}$ **C** $\dfrac{-e^2}{8\pi\varepsilon_0 a}$ **D** $\dfrac{e^2}{2\pi\varepsilon_0 a}$

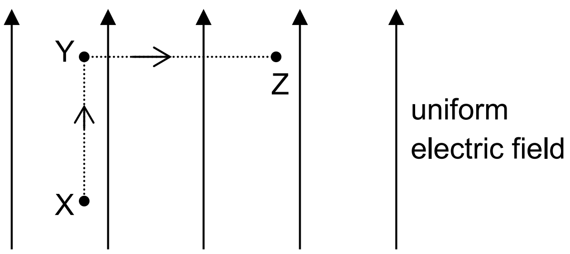
16 A small, charged ball of mass *m* is suspended by a light string in a uniform electric field of field strength *E*. The ball is at rest when the string makes an angle θ with the vertical.



What is the charge on the ball?

A $\dfrac{-mg}{E}$ **B** $\dfrac{mg}{E}$ **C** $\dfrac{-mg}{E}\tan\theta$ **D** $\dfrac{mg}{E}\tan\theta$

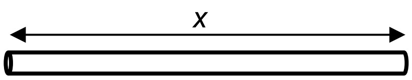
17 X, Y and Z are points in a uniform electric field of field strength $5.0\ \text{V m}^{-1}$. A point charge of charge $+3.0\ \text{C}$ is initially at rest at X. It is then moved 3.0 m parallel to the field lines to Y and 4.0 m perpendicular to the field lines to Z.



What is the decrease in the electric potential energy of the point charge?

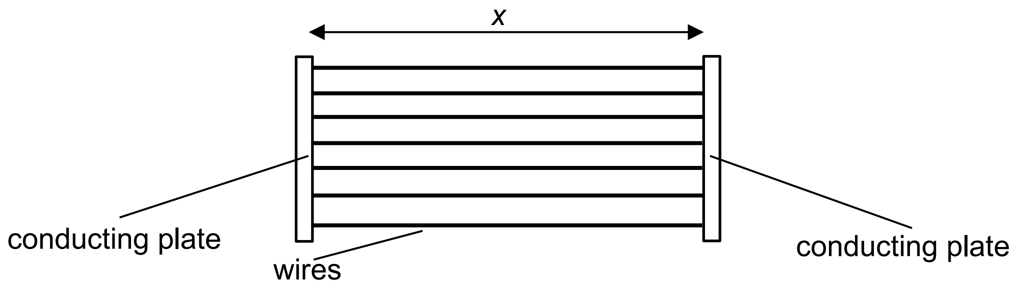
A 5.0 J **B** 15.0 J **C** 45.0 J **D** 60.0 J

18 A manufacturer has two pieces of constantan of the same volume. The first piece is made into a cylindrical resistor P of length x.



resistor P

The second piece is made into uniform wires each of length x which are connected to two conducting plates of negligible resistance to form a resistor Q.



resistor Q

Electrical current flows along the length x of both resistors.

How do the electrical resistances of P and Q compare?

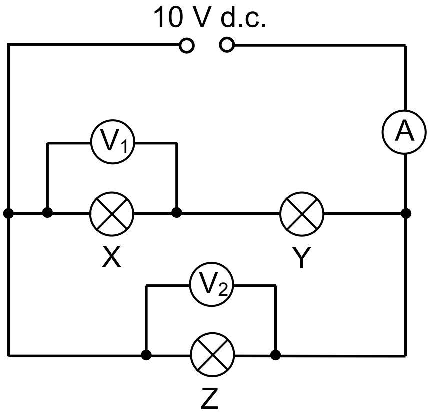
A P has a larger resistance than Q.

B Q has a larger resistance than P.

C P and Q have the same resistance.

D Q may have a larger or smaller resistance than P, depending on the number of wires made.

19 A 10 V d.c. power supply is connected to three identical light bulbs X, Y and Z. An ideal ammeter and two ideal voltmeters are connected to the circuit as shown. The ammeter has a reading of 1 A initially.



The filament of bulb X becomes broken.

What are the new readings of the voltmeters and the ammeter?

	voltmeter V_1	voltmeter V_2	ammeter
A	10 V	10 V	smaller than 1 A
B	10 V	10 V	larger than 1 A
C	0 V	5 V	1 A
D	0 V	10 V	1 A

20 A small compass placed horizontally points North due to the horizontal component of the magnetic flux density of the Earth's magnetic field. The horizontal component of the magnetic flux density at this location is 3.5×10^{-5} T.

A vertical wire is placed 9.0 mm due South of the compass. When a current of 3.0 A flows downwards through the wire, the compass needle deflects.

What is the angle and direction of the deflection?

A 28° East of North

B 28° West of North

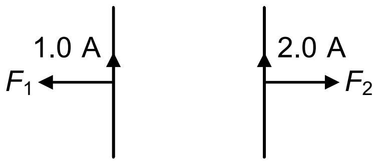
C 62° East of North

D 62° West of North

21 Two long, straight, parallel wires carry currents of 1.0 A and 2.0 A.

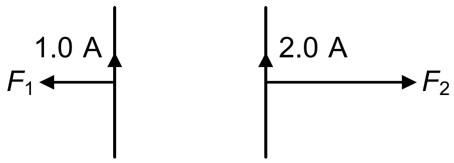
Which diagram shows the directions and relative magnitudes F_1 and F_2 of the forces per unit length on each wire?

A



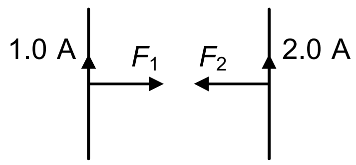
$F_2 = F_1$

B



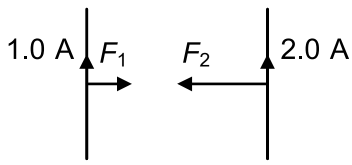
$F_2 = 2F_1$

C



$F_2 = F_1$

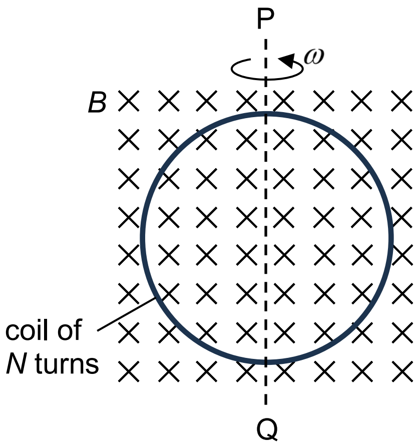
D

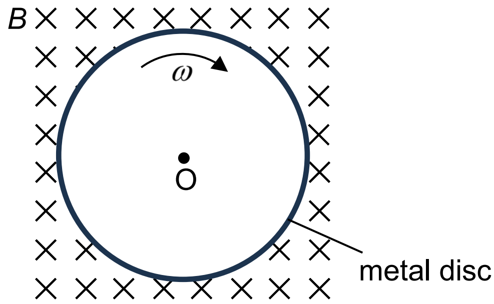


$F_2 = 2F_1$

22 A coil of N turns and area A is being rotated at a constant angular speed ω about the vertical axis PQ in a uniform magnetic field of flux density B.

In another setup, a metal disc of area A is being rotated at the same constant angular speed ω about its centre O in a uniform magnetic field of the same flux density B.

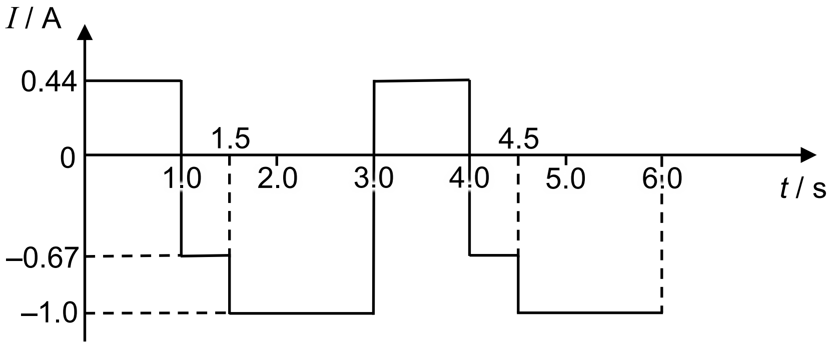




What is the ratio of the peak e.m.f. induced in the coil to the e.m.f. induced across the centre O and the edge of the metal disc?

A N **B** $\dfrac{2\pi N}{\omega}$ **C** $2\pi N$ **D** $4\pi N$

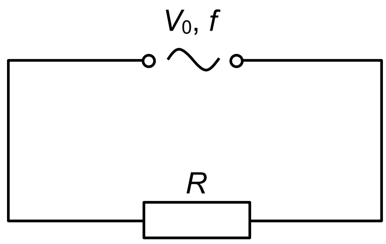
23 The graph shows the variation with time t of an alternating current I in a circuit.



What is the r.m.s. current in the circuit?

A 0.64 A **B** 0.80 A **C** 1.4 A **D** 1.9 A

24 A resistor of resistance R is connected to a sinusoidal a.c. power supply of peak voltage V_0 and frequency f. The mean power dissipated in the resistor is P.



Which of the following changes to the circuit will decrease the mean power dissipated in the resistor to $\frac{P}{4}$?

A Increase the resistance of the resistor to $4R$.

B Decrease the peak voltage of the a.c. power supply to $\frac{V_0}{4}$.

C Decrease the frequency of the a.c. power supply to $\frac{f}{4}$.

D Connect an ideal diode in series with the a.c. power supply and the resistor.

25 Which statement about the photoelectric effect is correct?

A Photoelectrons are not emitted as long as intensity of illumination is low.

B Doubling the frequency of the incident radiation will double the stopping potential.

C For a particular clean metal surface, there will be a minimum wavelength below which no emission of photoelectrons will occur.

D Increasing the intensity of the incident radiation increases the photocurrent.

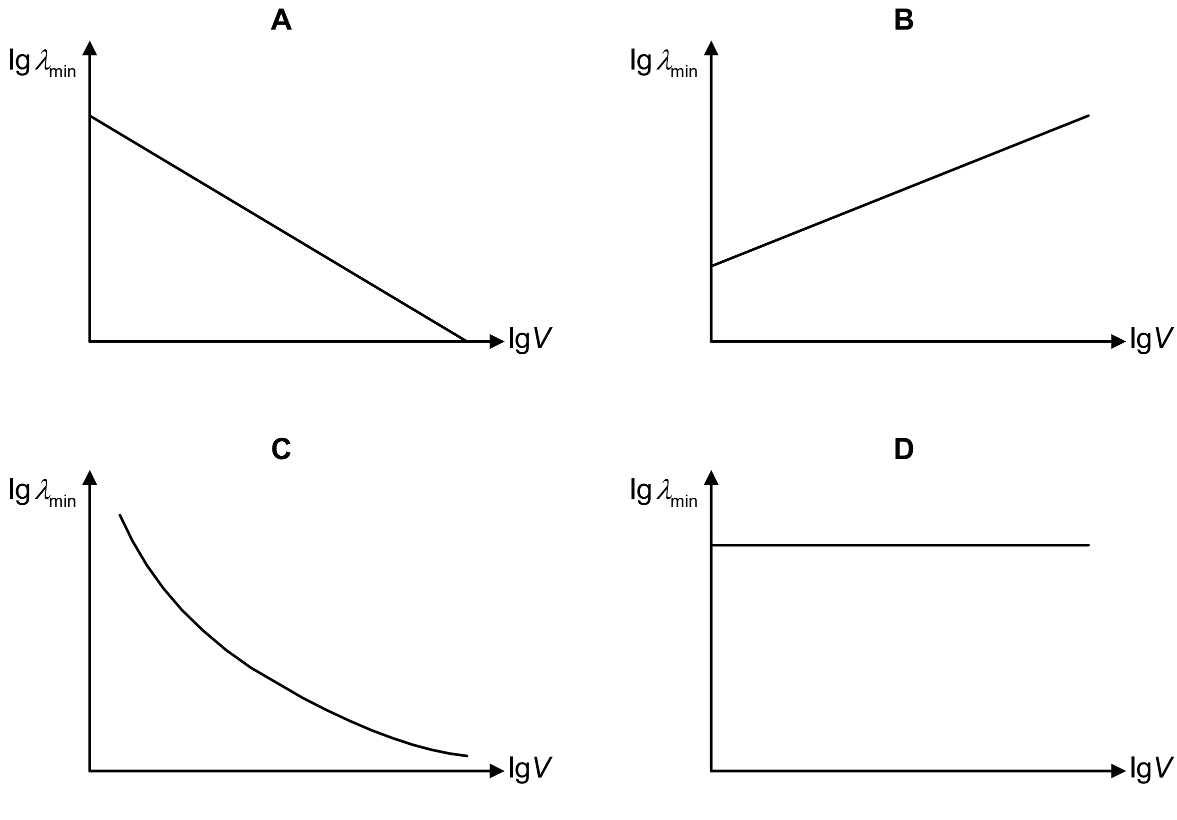
26 The kinetic energy of an electron is five times that of a proton.

What is the ratio of the de Broglie wavelength of the electron to that of the proton?

A 0.052 **B** 0.20 **C** 0.44 **D** 19

27 Electrons are accelerated from rest by a potential difference V. They bombard a metal target to produce X-rays. On the resulting X-ray spectrum, λ_{min} is the shortest wavelength observed.

Which graph shows the variation with $\lg V$ of $\lg \lambda_{min}$?



28 The nucleus X has the notation ${}^{P}_{Q}\mathrm{X}$. The mass defect of this nucleus is Δm.

What is the binding energy per nucleon of the nucleus X, where c is the speed of light in free space?

A $c^2\Delta m$ **B** $Pc^2\Delta m$ **C** $\dfrac{c^2\Delta m}{P-Q}$ **D** $\dfrac{c^2\Delta m}{P}$

29 A stationary uranium-238 $\left({}^{238}\mathrm{U}\right)$ nucleus decays by emitting an α-particle and a thorium-234 $\left({}^{234}\mathrm{Th}\right)$ nucleus, generating a total kinetic energy of E.

What is the kinetic energy of the α-particle?

A $\dfrac{E}{2}$

B E

C slightly less than $\dfrac{E}{2}$

D slightly less than E

30 The average value of the background radiation is recorded as 18 counts per minute by a detector of ionising radiation.

A radioactive source is placed close to the detector, and the detector records an average of 602 counts per minute.

What is the average counts per minute that will be recorded by the detector after three half-lives of the radioactive source?

A 73 **B** 75 **C** 91 **D** 93

END OF SECTION A